Earth-Science Reviews 218 (2021) 103679

Contents lists available at ScienceDirect

Earth-Science Reviews

journal homepage: www.elsevier.com/locate/earscirev

Revisiting the Permian seawater $^{87}Sr/^{86}Sr$ record: New perspectives from brachiopod proxy data and stochastic oceanic box models

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ARTICLE INFO

Keywords:
Permian
Strontium isotopes
Continental weathering
Hydrothermal
Brachiopods
Mass balance model

ABSTRACT

The Permian Period is punctuated by Earth system changes unlike any other in geological history. The start of the Permian witnessed the termination of the Late Paleozoic Ice Age, followed by the climatic transition from icehouse to greenhouse conditions. The Guadalupian-Lopingian (Middle-Late Permian) was characterized by two biocrises associated to volcanisms: (i) the end-Guadalupian crisis and (ii) the end-Permian mass extinction. Seawater Sr isotope ($^{87}Sr/^{86}Sr$) records can shed light on the evolution of the Permian Earth system. The Permian $^{87}Sr/^{86}Sr$ record suffers from a number of issues including low resolution and potential diagenetic alteration. In this paper, we summarize the existing Permian $^{87}Sr/^{86}Sr$ records and focus on the current challenges. We also present a new, high-resolution Permian $^{87}Sr/^{86}Sr$ curve derived from pristine brachiopod shells based on data resulting from careful diagenetic screening. Our new record shows that the $^{87}Sr/^{86}Sr$ of seawater decreased continuously from the earliest Permian to the middle Capitanian (late Guadalupian), with the lowest ratio of 0.706832 registered in the *Colania douvillei-Kahlerina pachiytheca* Zone. Subsequently, $^{87}Sr/^{86}Sr$ ratios increased from the late Capitanian to the latest Permian and reached a ratio of 0.707167 just 0.8 m below the first occurrence of the *Hindeodus parvus*. We employed a stochastic oceanic box model to explore the potential drivers of the Permian seawater Sr isotope record. Our results support that changes in the hydrothermal input, rather than changes in continental weathering intensity, are the most likely controlling factor for the observed variations in Permian seawater $^{87}Sr/^{86}Sr$. Therefore, we suggest that the marine hydrothermal system (and hence ocean basin dynamics and deep-sea temperatures) may have been the driver of the pronounced decreasing $^{87}Sr/^{86}Sr$ trend across the Permian.

1. Introduction

The Permian (~ 299–252 Ma) experienced multiple severe environmental perturbations. The Cisuralian (Early Permian) witnessed the termination of the Late Paleozoic glaciation (Chen et al., 2013; Montañez and Poulsen, 2013). The Guadalupian-Lopingian (Middle-Late Permian) captured two mass extinction events (e.g., Erwin, 2006; Shen et al., 2019), several global-scale volcanic events (Burgess et al., 2017; Chen and Xu, 2021; Shellnutt et al., 2014; Wang et al., 2015; Zhong et al., 2020), and significant sea-level rise and fall (e.g., Haq and Schutter, 2008). The forcing(s) responsible for the climatic transition from the early Permian icehouse (Isbell et al., 2003b; Montañez and Poulsen, 2013) to the late Permian hothouse have received much attention in the past several decades (Brand et al., 2012b; Chen et al.,

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https://doi.org/10.1016/j.earscirev.2021.103679
Received 31 January 2021; Received in revised form 28 April 2021; Accepted 10 May 2021
Available online 12 May 2021

2016; Isbell et al., 2003a; Joachimski et al., 2012; Montañez and Poulsen, 2013). The icehouse-to-greenhouse transition and the subsequent rapid climatic warming across the Permian-Triassic boundary (PTB) have been regarded as a potential analogue to the modern environmental and biotic changes in response to current global warming (Tierney et al., 2020; Westerhold et al., 2020; Zachos et al., 2001).

It has been recognized that continental weathering dynamics played a crucial role in regulating global climate and controlling marine redox chemistry changes (e.g., Zhang et al., 2018, 2020a, 2020b). Many studies have attempted to reconstruct Permian paleoenvironmental history, specifically weathering on continents and hydrothermal dynamics, by using the strontium isotope (^{87}Sr/^{86}Sr) system (Kani et al., 2008, 2013; Korte et al., 2006; Martin and Macdougall, 1995; McArthur et al., 2012). From the Sr cycle perspective, the ^{87}Sr/^{86}Sr of seawater is at present controlled mainly by the input of sources with distinct Sr isotopic compositions: (i) the high-temperature hydrothermal flux delivering unradiogenic, mantle-like ^{87}Sr/^{86}Sr ratios (~0.7037; Li and Elderfield, 2013), (ii) temperature-dependent, low-temperature hydrothermal flux carrying a relatively high Sr isotopic compositions (~0.7084; Coogan and Dosso, 2015), and (iii) the continental flux comprising the most radiogenic Sr input to the oceans (~0.71107; Peucker-Ehrenbrink and Fiske, 2019). Given the distinct isotopic composition of these fluxes, seawater ^{87}Sr/^{86}Sr variations may track shifts in continental weathering as well as high- and low-temperature hydrothermal inputs. Marine carbonates, evaporites, as well as skeletal hard parts have been treated as valuable archives for seawater ^{87}Sr/^{86}Sr ratios (Burke et al., 1982; McArthur et al., 2020; Palmer and Edmond, 1989; see discussion in Immenhauser et al., 2016). Despite the fact that coastal process may shift local ^{87}Sr/^{86}Sr records (e.g., El Meknassi et al., 2020; Holmden et al., 1997; Ingram and Sloan, 1992), the much longer residence time of Sr in Earth's ocean (~10^6 yr) permits a global-scale correlation on ^{87}Sr/^{86}Sr (Brand et al., 2003; El Meknassi et al., 2018; Elderfield, 1986; Pearce et al., 2015; Zaky et al., 2019).

The current Permian seawater ^{87}Sr/^{86}Sr record depicts a significantly decreasing trend from the latest Carboniferous to the late Guadalupian, followed by an increase towards the latest Permian (Burke et al., 1982; Korte et al., 2006; Peterman et al., 1970; Veizer and Compston, 1974; Wang et al., 2018). The sharp drop from the earliest Permian to the Middle Permian is recognized as the largest continuous decline in marine ^{87}Sr/^{86}Sr ratios of the Phanerozoic (Korte and Ullmann, 2016; McArthur et al., 2020). The ^{87}Sr/^{86}Sr ratios in the late-Guadalupian registered the lowest values in the Paleozoic (Kani et al., 2008, 2013, 2018). A number of hypotheses have been put forward to explain the decline in seawater ^{87}Sr/^{86}Sr from the Early to Middle Permian. These include changes in the riverine Sr flux (Martin and Macdougall, 1995), reduced continental weathering (Korte et al., 2006), and increased input of the hydrothermal flux (Wang et al., 2018). Many of these interpretations, however, are based on the qualitative theory of the Sr isotope system without quantitative assessment.

A significant number of studies have focused on geologically short intervals surrounding the Guadalupian-Lopingian boundary (GLB, ~ 259.1 Ma; Kani et al., 2008, 2013, 2018; Liu et al., 2013) and the PTB (~ 251.9 Ma; Brand et al., 2012b; Cao et al., 2009; Dudás et al., 2017; Garbelli et al., 2016; Huang et al., 2008; Martin and Macdougall, 1995; Sedlacek et al., 2014; Song et al., 2015), but interpretations remain controversial. Some of the debates center around (i) the stratigraphic position of the lowest ^{87}Sr/^{86}Sr ratios (Kani et al., 2008, 2013, 2018) and (ii) the rates of change in ^{87}Sr/^{86}Sr ratios at different Permian substages (Cao et al., 2009; Dudás et al., 2017; Huang et al., 2008; Song et al., 2015). These differences potentially result in diverging interpretations of geologic events and inaccurate stratigraphic correlations.

These inconsistencies may arise from the different archives used for ancient seawater ^{87}Sr/^{86}Sr reconstruction. Commonly-used archives include: (i) evaporites (Brookins, 1988), (ii) bulk micritic limestones (Cao et al., 2009; Denison et al., 1994; Huang et al., 2008; Kani et al., 2008, 2013, 2018; Sedlacek et al., 2014; Wang et al., 2018), (iii) conodonts (Dudás et al., 2017; Martin and Macdougall, 1995; Song et al., 2015), and (iv) brachiopod shells (Brand et al., 2012b; Garbelli et al., 2016, 2019; Korte et al., 2006; Popp et al., 1986). Those different archives sometimes yield different ^{87}Sr/^{86}Sr ratios for the same time interval. In particular, this is the case for the PTB when comparing results from conodonts, brachiopod shells, and micritic limestones (Cao et al., 2009; Korte et al., 2006; Song et al., 2015). Various diagenesis-screening methods and criteria could also lead to different records and interpretations. For example, McArthur et al. (2012) showed significantly different interpretations when applying different diagenesis filters on the ^{87}Sr/^{86}Sr record. Bulk limestone may be prone to secondary mineral contamination, driving the Sr isotopic composition more radiogenic (Bellefroid et al., 2018), whereas conodont and oyster samples have a higher possibility of being altered (McArthur et al., 2012).

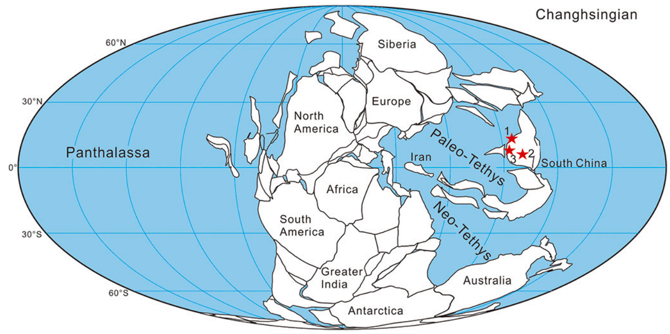

Fig. 1. Paleogeography maps showing position of the studied sections during the Changhsingian, 1. Xikou section; 2. Bianping section; 3. Shangsi section.

Calcitic brachiopod shells are considered as one of the best archives for capturing contemporaneous seawater ^{87}Sr/^{86}Sr due to their resistance to diagenesis (Brand et al., 2011; Popp et al., 1986). Furthermore, analyses of modern brachiopod shells show little taxonomic or vital effects on ^{87}Sr/^{86}Sr (e.g., El Meknassi et al., 2018; Immenhauser et al., 2016; Zaky et al., 2019). However, a high-resolution Permian ^{87}Sr/^{86}Sr record based on this material has been hampered by the sporadic occurrences and collections of brachiopods in these strata. Consequently, the objectives of this study are: 1) to provide a comprehensive summary and evaluation of the published Permian ^{87}Sr/^{86}Sr data, 2) to present a new, high-resolution ^{87}Sr/^{86}Sr ratio record based on well-preserved brachiopod shells, 3) to compare new brachiopod data with the previously published ^{87}Sr/^{86}Sr records, and 4) to quantitatively test the potential controlling factors affecting the Permian ^{87}Sr/^{86}Sr ratios with a stochastic oceanic Sr box model.

2. Geological background

Brachiopods were collected from three sections: the Xikou section (33.23° N, 109.38° E) at Zhen'an county in Shaanxi province (Northwestern China), the Bianping section (25.417°N, 107.767°E) at Ziyun county in Guizhou province (South China), and the Shangsi section (32.533°N, 106.75°E) at Guangyuan county in Sichuan province (South China). The Permian strata are well exposed at the Xikou section, containing 90% carbonate sequences with abundant fusulinids, brachiopods, and corals. From this section, brachiopods were collected from more than 300 horizons. The lower part of the Xikou section, however, Cisuralian in age, contains few brachiopods. Thus, Cisuralian's brachiopods were collected from the Bianping (Asselian, Sakmarian, and Artinskian) and Shangsi (Kungurian) sections. During the Permian, the three sections were near the paleo-equator (Fig. 1) (Dong et al., 2015) and can be correlated by fusulinid and conodont biostratigraphy (Fig. 2).

2.1. Xikou section

The Permian carbonate sequences of the Xikou section were divided

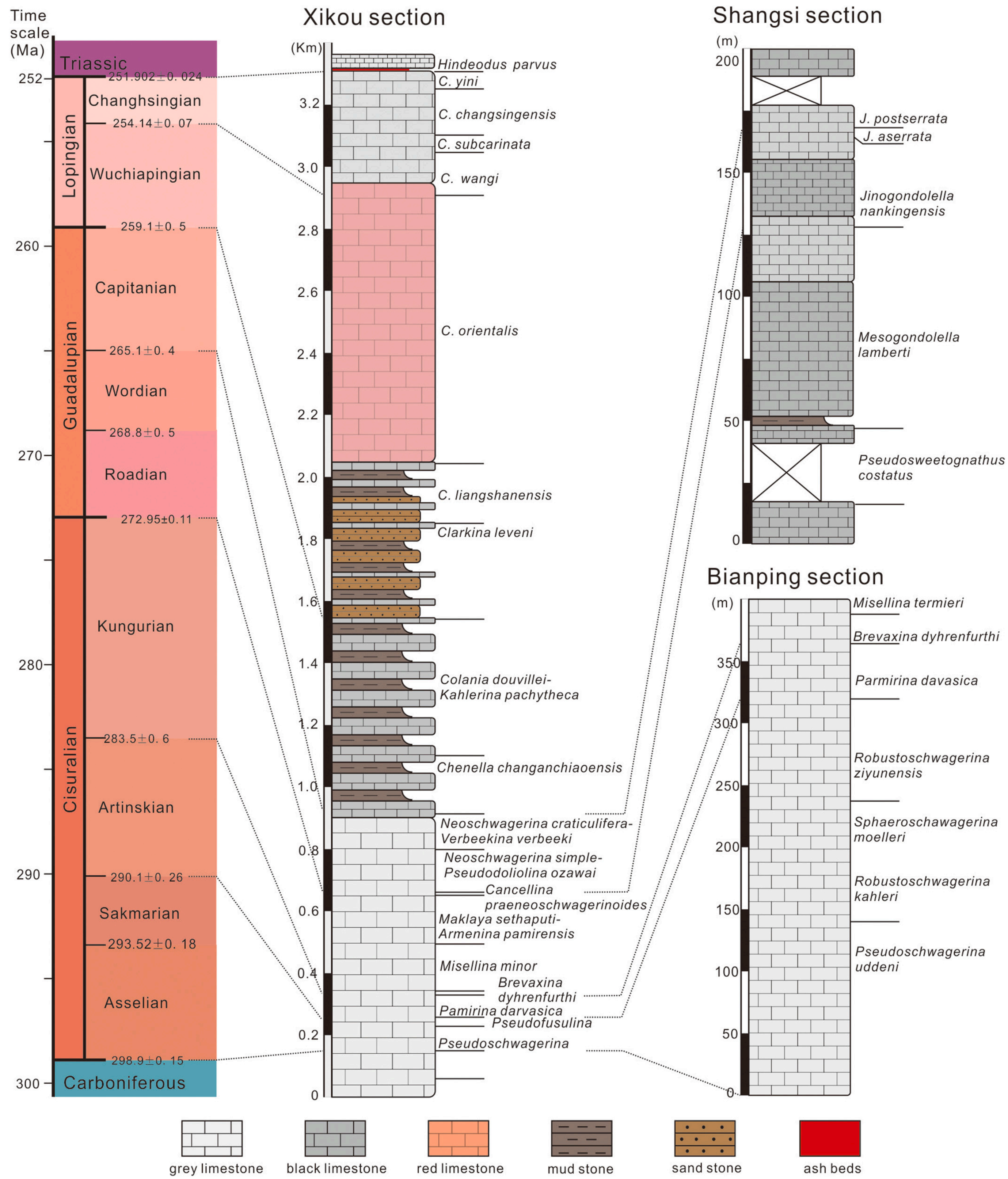

Fig. 2. Stratigraphic columns with conodont, fusulinid zones and geochronologic ages. The ages are from the latest version of the Permian timescale (Shen et al., 2019). The biozones in the Xikou section are unpublished data from Zhang YC et al. and Yuan DX et al., the Lopingian strata are constrained by conodont zones and the Cisuralian and Guadalupian strata are divided by fusulinid zones. The fusulinid zones of the Bianping section are from Xiao et al. (1986). The conodont zones of the Shangsi section are from Yuan (2015, unpublished thesis).

into eight formations by Ding et al. (1989). In ascending order, these are the Sanlichong, Shimenya, Yazi, Wulipo, Shuixiakou, Xikou, Yundoutan, and Longdongchuan formations. The Sanlichong to Wulipo formations developed thickly-bedded bioclastic limestones containing abundant fusulinids and were assigned to the Cisuralian to lower Guadalupian stages. The occurrence of *Pseudoshwagerina* defines the boundary between the Carboniferous and the Permian periods. The occurrences of *Pseudofusulina*, *Pamirina*, *Brevaxina*, *Neoschwagerina*, and *Presumatrina* define the base of the Sakmarian, Artinskian, Kungurian, Roadian, and Wordian stages, respectively. The Shuixiakou Formation contains middle- to thickly-bedded limestone interbedded with siltstones, which belong to the Capitanian stage based on the occurrence of *Chenella*. The Xikou Formation is composed of sandstone interbedded with limestone and siltstone; it can be differentiated from the Shuixiakou Formation by the appearance of siliciclastic beds, situated above an ash bed that can be correlated with the Emeishan volcanism. The disappearance of the bigger fusulinids, replaced by the smaller fusulinids such as *Kahlerina*, *Codonofusiella,* and *Reichelina,* suggested this formation can be ascribed to Wuchiapingian. The Yundoutan Formation is characterized by thickly bedded to massive reddish limestones, with abundant brachiopods, corals, crinoids, and sponges. The Longdongchuan Formation contains thickly bedded to massive white limestones. The first occurrences of *Clarkina wangi* and *Hindeodus parvus* were recognized in the Longdongchuan Formation, which can be defined as the base of the Changhsingian and Induan stages, respectively.

2.2. Bianping section

The Permian sequence in the Bianping section contains thickly-bedded bioclastic limestones, with many fusulinids, brachiopods, corals, bivalves, and gastropods. Xiao et al. (1986) reported detailed fusulinid biostratigraphy, from *Pseudoschwagerina uddeni* to *Misellina claudiae* Zones. Therefore, the brachiopod shells collected from this section can be correlated with all other sections in the Paleotethys by means fusulinid biostratigraphy (Xiao et al., 1986; Shen et al., 2019).

2.3. Shangsi section

The Shangsi section contains the sequence from the upper Cisuralian Chihsia to the Lower Triassic Feixianguan formations and was proposed as the candidate Global Stratotype Section and Point (GSSP) for PTB (Lai et al., 1996). Previous studies focused on biostratigraphy suggesting a Guadalupian to Early Triassic in age for this section. Yuan (2015) analyzed conodonts from the Chihsia Formation base, and reported several conodont biozones, including the *Pseudosweetognathus costatus* to *Jinogondolella postserrata* zones (Fig. 2) and confirmed these strata can be correlated to the Kungurian to Capitanian stages. Brachiopod shells were collected from the Chihsia Formation in this section. The Chihsia Formation is differentiated from the Maping Formation by the Liangshan member, a suite of black shale interbedded with limestones in the Shangsi section.

The age information for individual samples was estimated according to the geochronological framework published by Shen et al. (2019), assuming consistent sedimentation rates within substages.

3. Materials and methods

3.1. Materials

Rhynchonelliformean brachiopods and associated whole rock samples from the Moscovian to the latest Changhsingian were collected from the Xikou, Bianping, and Shangsi sections. A total of 277 brachiopod shells were collected and screened for $^{87}Sr/^{86}Sr$ analysis. Brachiopods in the Bianping and Shangsi sections are all spiriferids, while productids and athyrids are also present in the Xikou section (Appendix A).

3.2. Scanning Electron and Cathodoluminescence Microscopy

Multiple screening processes were employed to evaluate the preservation and diagenesis status of our samples (Brand et al., 2012a; Garbelli et al., 2012, 2019; Ullmann and Korte, 2015). These screening methods include microstructure analysis under the Scanning Electron Microscope (SEM), cathodoluminescent microscopy, trace elements, and isotope analyses (Brand, 2004; Grossman et al., 1991; Korte et al., 2006; Popp et al., 1986; Ullmann and Korte, 2015). The microscopic methods utilized in this study are explained in detail in Garbelli et al. (2019) and Wang et al. (2020).

The shells were cut along their longitudinal axis, embedded in epoxy resin, polished and etched with 5% HCl for 8–10 s, coated with gold, and scanned by LEO 1450 VP located at the Nanjing Institute of Geology and Palaeontology (NIGP), Chinese Academy of Sciences (CAS). The shells were evaluated as “well-preserved” if their fabric retains the original microstructure of different layers. If the secondary layer shows poor preservation, but the tertiary layer is well preserved, the shells were deemed as “partially preserved”. The shells were considered “altered” when their fabric lost their original structure (Casella et al., 2018; Garbelli et al., 2019).

The shells were then cut into thin sections and analyzed by cathodoluminescent microscopy at NIGP, with a cold cathode luminescence microscope (Nuclide ELM2) operating at 10 kV with a beam current 5–7 mA. The exposure time to the electronic beam (before taking the photo) was limited to 15–30 s. Photographic exposure time was set to ~2 s with a ZEISS Colour 506 Camera. The shells were classified as non-luminescent if they appear dull to black, weakly luminescent when they yield luminescent microfractures, and bright luminescent when the shells display orange luminescence.

Combining microstructural and cathodoluminescence evaluation, shells were classified as preserved if they bear well-preserved fabric and

Table 1
Strontium mass balance model parameters.

Symbol	Description	Initial model ranges	1σ for step 2 of model	Modern value	Source
F_{riv}	Riverine flux	10–190 × 10^9 mol yr^{-1}	10%	47.6 × 10^9 mol yr^{-1}	¶
R_{riv}	Riverine isotopic composition	0.7025–0.75	0.0005	0.71107	¶
F_{lowT}	Low-temperature hydrothermal flux	2.5–40 × 10^9 mol yr^{-1}	10%	10 × 10^9 mol yr^{-1}	#
R_{lowT}	Low-temperature hydrothermal isotopic composition	0.7025–0.7084	0.0005	0.7084	#
F_{highT}	High-temperature hydrothermal flux	2–35 × 10^9 mol yr^{-1}	10%	8.4 × 10^9 mol yr^{-1}	||
R_{highT}	High-temperature hydrothermal isotopic composition	0.703–0.704	0.0001	0.7037	||
F_{dia}	Diagenetic flux	Kept constant		3.4 × 10^9 mol yr^{-1}	||
R_{dia}	Diagenetic isotopic composition	Kept constant		0.7084	||

Modern strontium flux and isotopic composition estimates utilized as reference for parameterization for the box model adapted from ¶ Peucker-Ehrenbrink and Fiske, 2019, # Coogan and Dosso, 2015, and || Li and Elderfield, 2013 and references therein.

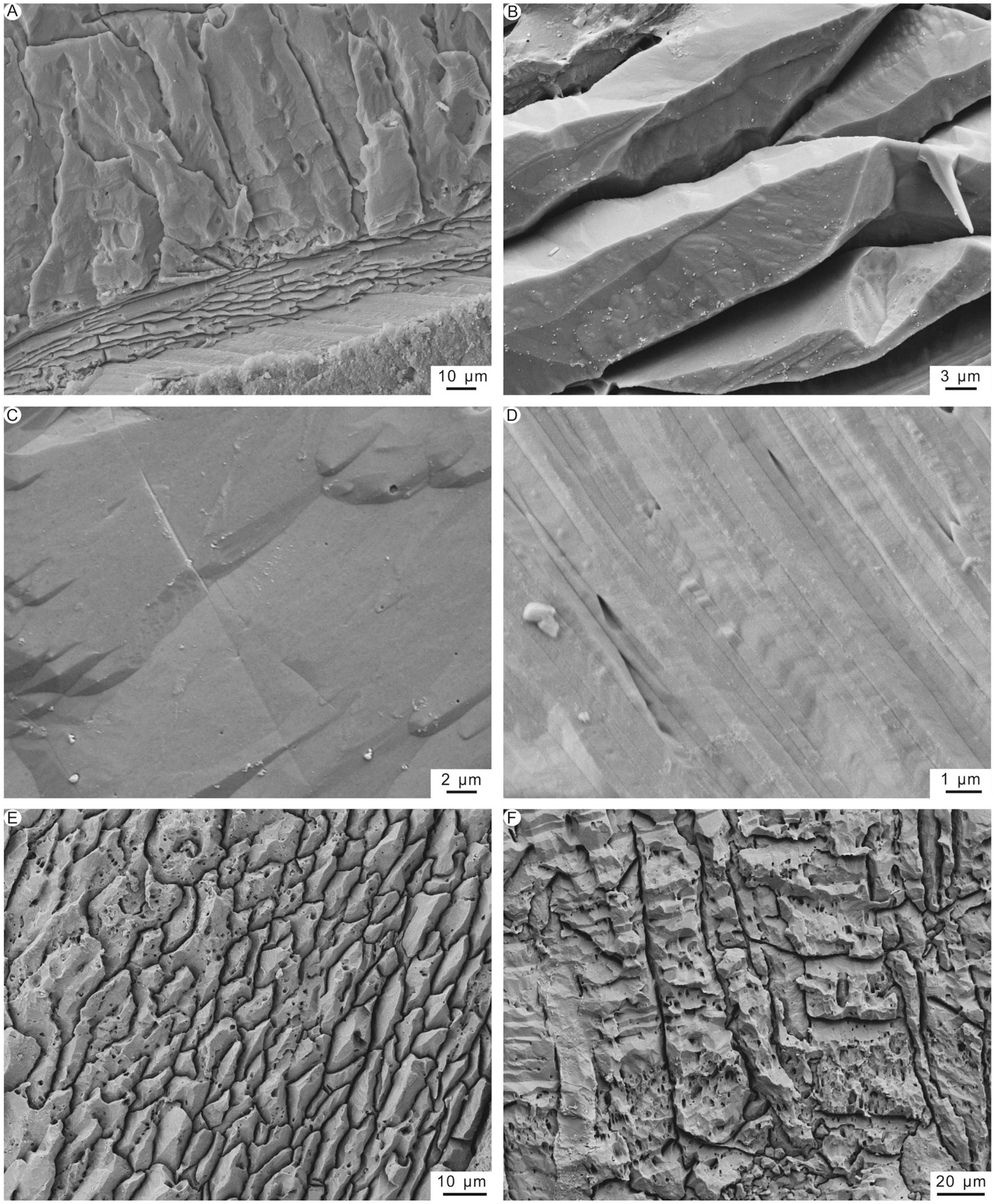

Fig. 3. Microstructure of shells. A. Specimen XK-8, *Permophricodothyris* sp., longitudinal section of the secondary fibrous and tertiary columnar layers, showing good preservation; B. Specimen BP-59, *Martinia* sp., longitudinal section showing well preserved fibrous layer; C. Specimen XK-178, *Permophricodothyris* sp., columnar layer, showing good preservation; D. Specimen XK-62, Monticuliferinae sp. indet., well preserved laminar layer; E. Specimen XK-77, *Tongzihyris* sp., altered fibrous layer, showing partly amalgamated fibers; F. Specimen XK-175, *Permophricodothyris* sp., altered columnar layer with coarse surface.

are no luminescent. Well-preserved and partially preserved fabric, combined with slightly luminescent and non-luminescent characters, respectively, results in the assignment to partially preserved. In all other cases, the shells are considered altered and were excluded from the subsequent analytical steps.

3.3. Elemental analysis

The shells were cleaned following the protocol of Zaky et al. (2015) before micro-drilling. About 10 mg powders were micro-drilled under a binocular microscope. More than two different positions for each shell were collected, along with their associated sedimentary matrix (Appendix B). Dorsal valves, specialized regions such as the beak, shell with punctae, and hollow spines, and the shells' fractures were avoided. A subset of 436 brachiopod samples from 173 specimens, representing 140 horizons was drilled for trace elemental analysis. Additionally, 100 whole rock samples were also analyzed for their elemental concentration to better understand the preservation state of brachiopod shells.

For the brachiopod shells, 5 mg of the shell powders were dissolved in 5 ml of doubly distilled HNO_3 (Brand et al., 2012b) for 70 min to

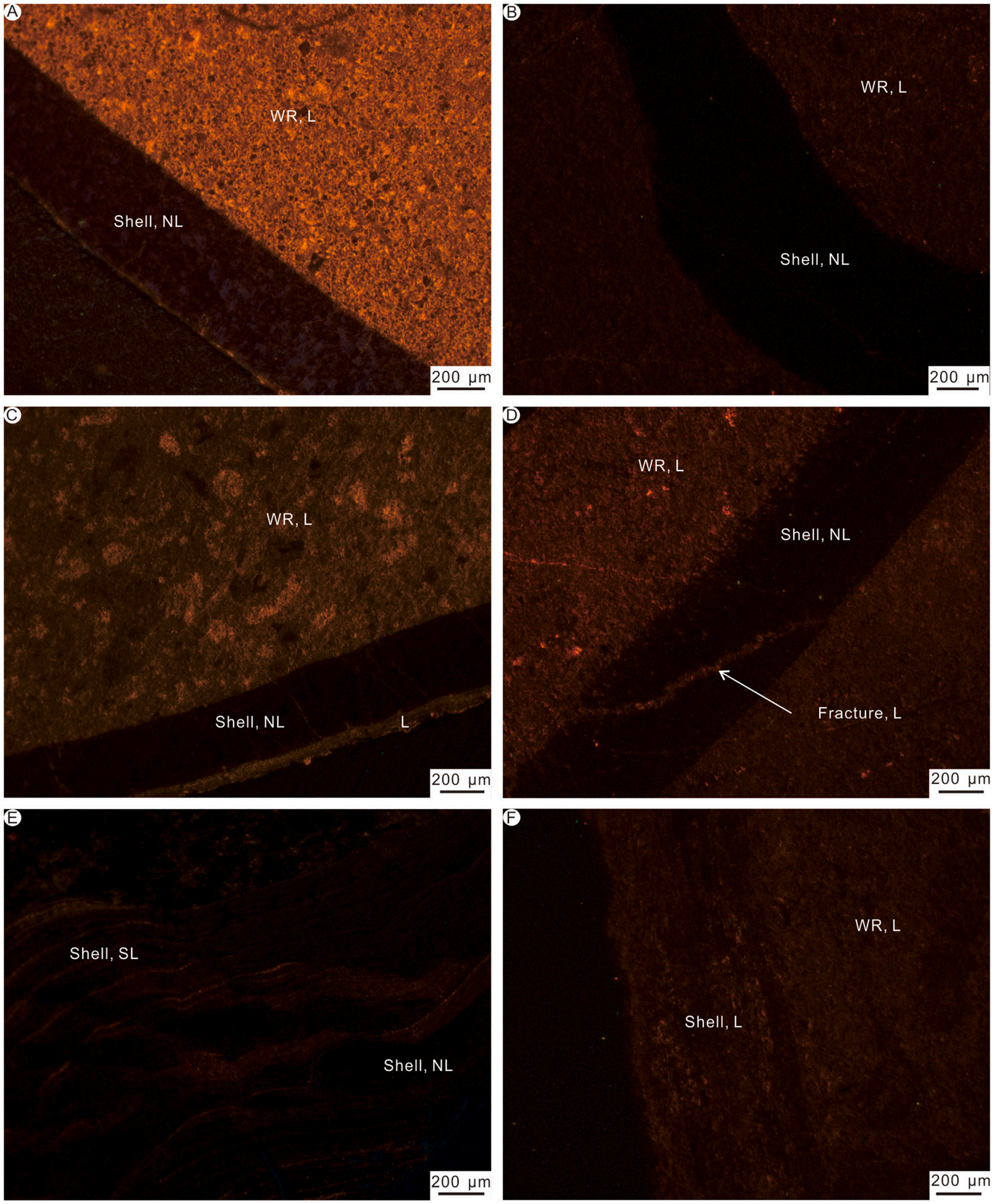

Fig. 4. Thin sections observed using cathodoluminesence microscopy. A. Specimen XK-13, *Permophricodothyris* sp., non-luminescent shell; B. Specimen BP-42, *Parachoristites* sp., non-luminescent shell; C. Specimen XK-17, *Permophricodothyris* sp., showing luminescent secondary layer and non-luminescent tertiary layer; D. Specimen XK-116, *Permophricodothyris* sp., non-luminescent shells with brightly luminescent fracture; E. Specimen XK-62, Monticuliferinae sp. indet., shell showing longitudinal slightly-luminescent fissures; F. Specimen XK-121 *Permophricodothyris* sp., shell showing brightly-luminescent characteristics. NL = non-luminescent; L = luminescent; WR = whole rock.

complete dissolution. For the sedimentary matrix samples, 50 mg powders were dissolved with 3 ml 1 M acetic acid in an ultrasonic water bath at 30 °C for 30 mins (Wang et al., 2018). Then the solution was left at room temperature for 12 h, centrifuged, and decanted three times. The supernatants were subsequently evaporated to dryness at 120 °C, and the non-dissolvable residues were dried and weighed to calculate the percent dissolution. Trace element concentrations were measured by a Spectroflame D inductively-coupled plasma at NIGP. Reference solutions prepared in-house were measured to calibrate the results during the analysis, and the reference standard JDo-1 (Imai et al., 1996) was used every ten samples. The reproducibility (2sd) was better than 2.7% for Ca, 5% for Fe, 4% for Mg, 3.8% for Mn, and 2.6% for Sr.

3.4. Carbon and oxygen isotope analyses

About 100 μg powders were reacted with ultrapure 102% HPO_3 at 70 °C in a Kiel IV device coupled with a Finnigan MAT 253 mass spectrometer at NIGP for the carbonate carbon isotope ($\delta^{13}C$) and oxygen

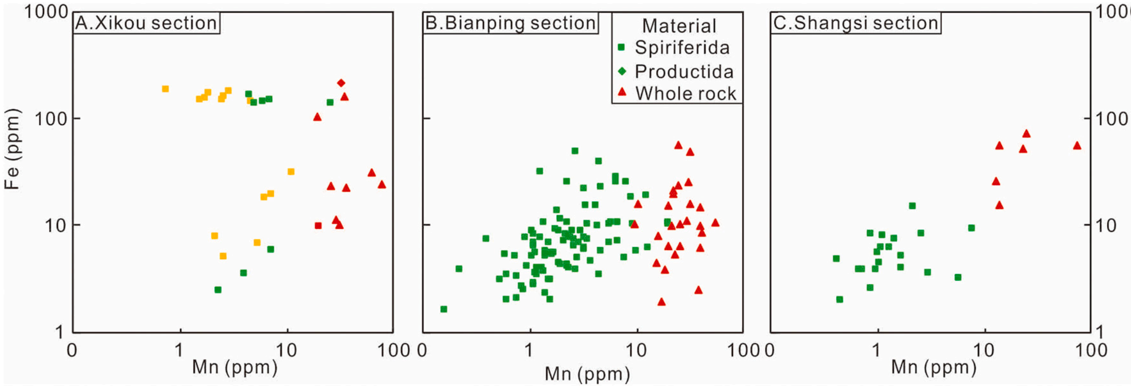

Fig. 5. Cross plot of Mn and Fe for different sections of Cisuralian. Green symbols represent preserved shells, yellow symbols represent partially preserved shells, and red symbols represent altered shells. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

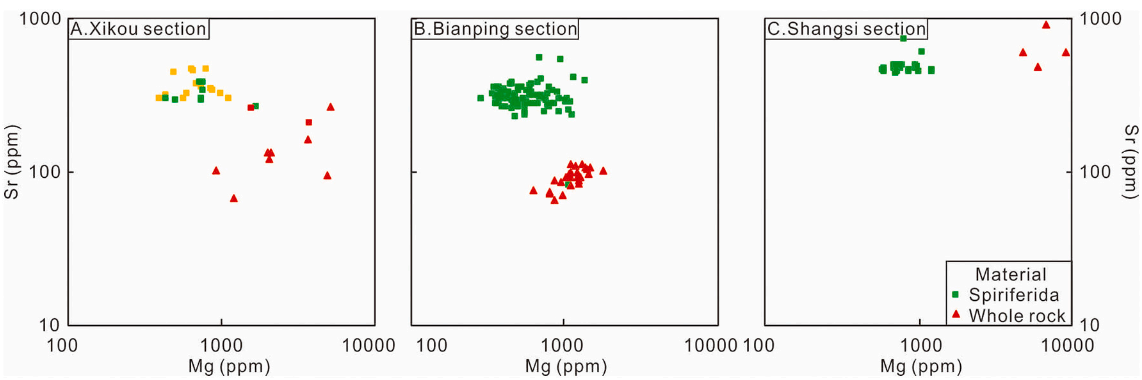

Fig. 6. Cross plot of Mg and Sr for different sections of Cisuralian. Green symbols represent preserved shells, yellow symbols represent partially preserved shells, and red symbols represent altered shells. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

isotope ($\delta^{18}O$) analyses. Each shell was micro-drilled at least two different positions, and some larger and thicker shells were drilled in 3–4 positions (Appendix B). A total of 455 $\delta^{13}C$ and $\delta^{18}O$ values of the brachiopod shells from 184 specimens were obtained and an additional 100 $\delta^{13}C$ and $\delta^{18}O$ analyses of the host matrix sediment were compiled. Standard GBW-04405 ($\delta^{13}C$: 0.57 ± 0.03‰, $\delta^{18}O$: −8.49 ± 0.13‰) was measured during the analyses, yielding 0.58 ± 0.03‰ for $\delta^{13}C$, −8.51 ± 0.02‰ (2sd, N = 24) for $\delta^{18}O$. The long-term precision and accuracy were better than 0.03‰ for $\delta^{13}C$ and 0.08‰ for $\delta^{18}O$. The $\delta^{13}C$ and $\delta^{18}O$ values are reported in per mil relative to V-PDB (Vienna Peedee belemnite).

3.5. Strontium isotope ratio analyses

A total of 184 $^{87}Sr/^{86}Sr$ ratios from 141 shells from 111 horizons were analyzed at Ruhr University, Bochum. Subsamples of 1 mg were dissolved with 6 M suprapure HCl for about 24 h at room temperature in closed PFA beakers and evaporated until dry. Dry samples were re-dissolved in 0.4 ml 3 M HNO_3. Strontium was isolated using an ion exchange resin (Sr-resin TRISKEM, basing on a crown-ether (4,4′(5′)-di-t-butylcyclohexano-18-crown-6) diluted in octanol) conditioned with 3 ml 3 M HNO_3 applying 1 ml distilled water. After evaporation, the dry sample material was treated with 1 ml of a 1:1 mixture of concentrated HNO_3 and H_2O_2 to remove organic remains and remnants of TRISKEM

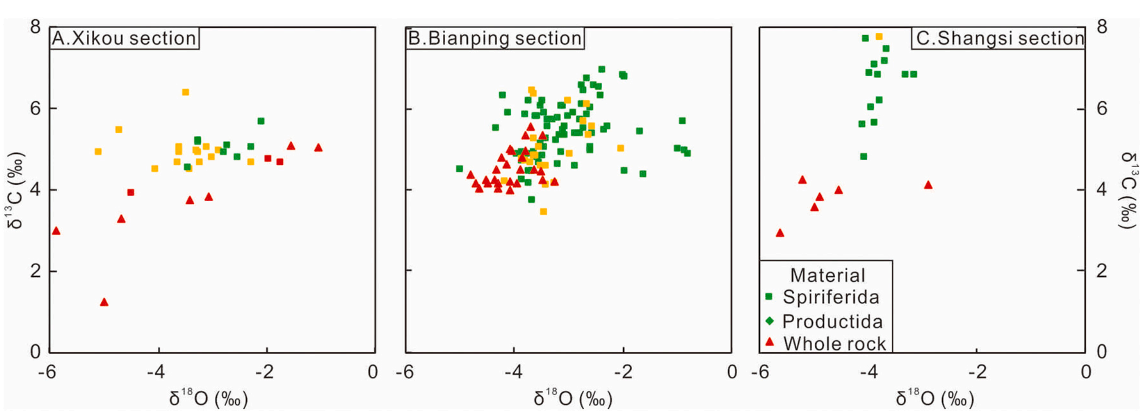

Fig. 7. Cross plot of $\delta^{13}C$ and $\delta^{18}O$ for different sections of Cisuralian. Green symbols represent preserved shells, yellow symbols represent partially preserved shells, and red symbols represent altered shells. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

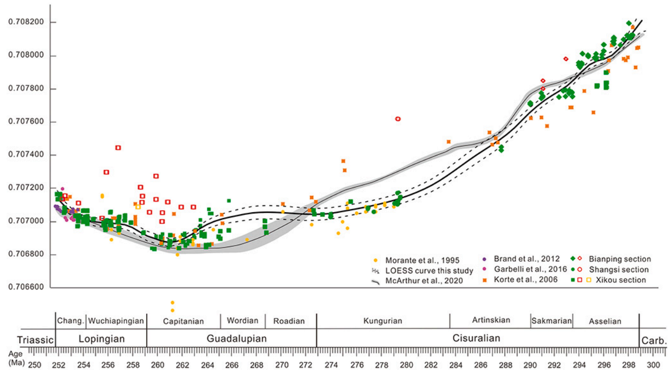

Fig. 8. $^{87}Sr/^{86}Sr$ data from this study using brachiopod shells of the Permian and previously published records calibrated to the latest updated geochronologic ages. The green solid symbols represent the preserved data, orange and red empty symbols represent partially preserved and altered data respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

resin and dried again. Finally, samples were converted to chloride form using 0.4 ml 6 M HCl for $^{87}Sr/^{86}Sr$ measurement. The solution was analyzed on a Spectromat Tibox (formerly Finnigan MAT 262) 7-collector solid-source mass spectrometer with a single Re filament in dynamic mode. The mean $^{87}Sr/^{86}Sr$ ratios of the USGS EN-1 and the NIST NBS987 standards were 0.709156 ± 0.000032 (2sd, N = 22), 0.710240 ± 0.000014 (2sd, N = 22), respectively. The $^{87}Sr/^{86}Sr$ ratios in this study were normalized to $^{86}Sr/^{88}Sr$ of 0.1194, and all the $^{87}Sr/^{86}Sr$ were corrected to NBS-987 $^{87}Sr/^{86}Sr$ ratios of 0.710247 (McArthur et al., 2001).

3.6. Strontium oceanic box model design

To explore the potential forcings on the Sr isotope system that may have generated the empirical $^{87}Sr/^{86}Sr$ seawater record, we leveraged a stochastic oceanic box model to track the effect of significant changes in Sr fluxes and associated isotopic compositions. More specifically, we utilized the following mass balance:

$$F_{in} = F_{riv} + F_{highT} + F_{lowT} + F_{dia} = F_{out} \tag{1}$$

where the input flux (F_{in}) is composed of the riverine (F_{riv}), high-temperature hydrothermal (F_{highT}), low-temperature hydrothermal (F_{lowT}), and diagenetic (F_{dia}) fluxes. On million-year timescales, the input of Sr to the global oceans is equal to the output flux (F_{out}). With this steady state assumption ($F_{in} = F_{out}$), we solved for the Sr isotopic composition of seawater at 1 myr time-steps:

$$R_{ocean} = \frac{F_{riv}{}^{*}R_{riv} + F_{highT}{}^{*}R_{highT} + F_{lowT}{}^{*}R_{lowT} + F_{dia}{}^{*}R_{dia}}{F_{riv} + F_{highT} + F_{lowT} + F_{dia}} \tag{2}$$

where R_{ocean} is the $^{87}Sr/^{86}Sr$ composition of seawater, R_x and F_x are the isotope ratio and flux (mol yr^{-1}) of the input sources (above), respectively.

Eq. 2 allows us to investigate changes in the fluxes and associated isotopic compositions such that the modeled solutions match the empirical record. Given that the Sr fluxes and isotopic compositions may have been different in the Permian relative to the present world, we used a stochastic approach to reduce bias in the model parameterizations and allow it to test for a large range of scenarios. The model determines the time-dependent solution space of the Sr isotope system using Monte Carlo resampling from random, uniformly distributed combinations of the flux and isotopic composition values (Table 1). At each time step, the model was run 5000 times, generating a large range of potential solutions. Those solutions were then filtered by using upper and lower filtering bounds set equal to the 95% confidence intervals derived from

Table 2
Summary of preservation state of brachiopod shells from the Guadalupian and Lopingian stages at the Xikou section.

	Shuixiakou Formation	Xikou Formation	Yundoutan Formation	Longdongchuan Formation
No. of shells	70	24	60	37
Altered shells	27	4	15	13
Shells with laminar secondary layers	29	1	12	1

the LOWESS-smoothed Sr isotope record (Fig. 9). The model was run a second time, using random, normally distributed combinations of the mean filtered results from the initial run, with uncertainty (Table 1). This second step allowed for a narrowing of the potential parameter space, given that our initial parameterization occupied such a large range. In summary, with this framework, we generated a large solution space by resampling at each time step, yet only evaluated the model outputs that match the empirical Sr record.

Traditionally, interpretations of the Permian Sr isotope record focused on shifts in continental weathering and/or high-temperature hydrothermal flux (Korte et al., 2006; Wang et al., 2018). Therefore, we allowed the riverine and high-temperature fluxes to vary by a factor of ~4 relative to modern values: 10 –190 × 10^9 mol yr^{-1} and 2 –35 × 10^9 mol yr^{-1}, respectively. We also allowed the isotopic composition of the riverine flux to vary between unradiogenic, mantle-like end member (0.7025) and a much more radiogenic end member (0.75), representing the potential of weathering variable lithologies. It has recently been shown that seawater chemistry may have played a major role in determining the isotopic composition of fluids emanating from high-temperature hydrothermal systems (Antonelli et al., 2017). In several of the scenarios that were tested, we allowed the isotopic composition of the high-temperature hydrothermal flux to vary between the range estimated from Antonelli et al. (2017) for the Permian (0.703–0.704). We also included parameterization for the low-temperature hydrothermal flux, which has a relatively strong temperature dependence (Coogan and Dosso, 2015; Coogan and Dosso, 2012). Given that global

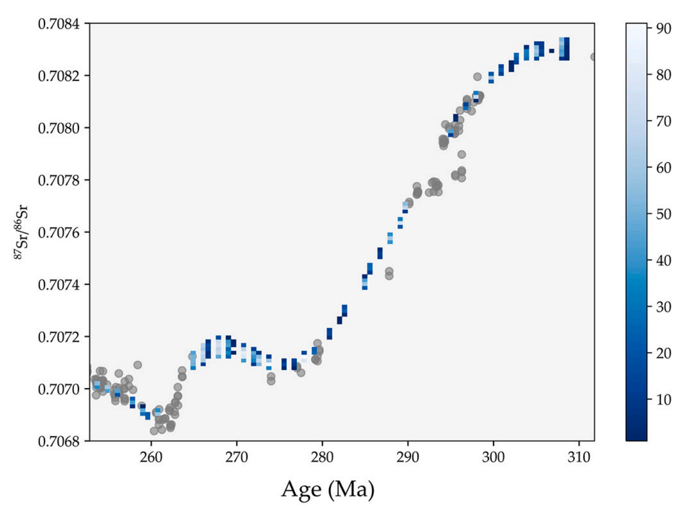

Fig. 9. Two-dimensional density heatmap of seawater strontium isotope mass balance results. Light blue colors indicate highest density (i.e., count) of successful model runs that match the empirically-determined, LOWESS-smoothed Sr isotope record using the 95% confidence intervals as filtering bounds for the stochastic model. The results presented here are from Scenario 1 (see section 4.4), and are similar for all scenarios. Grey circles represent Sr isotope data from this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

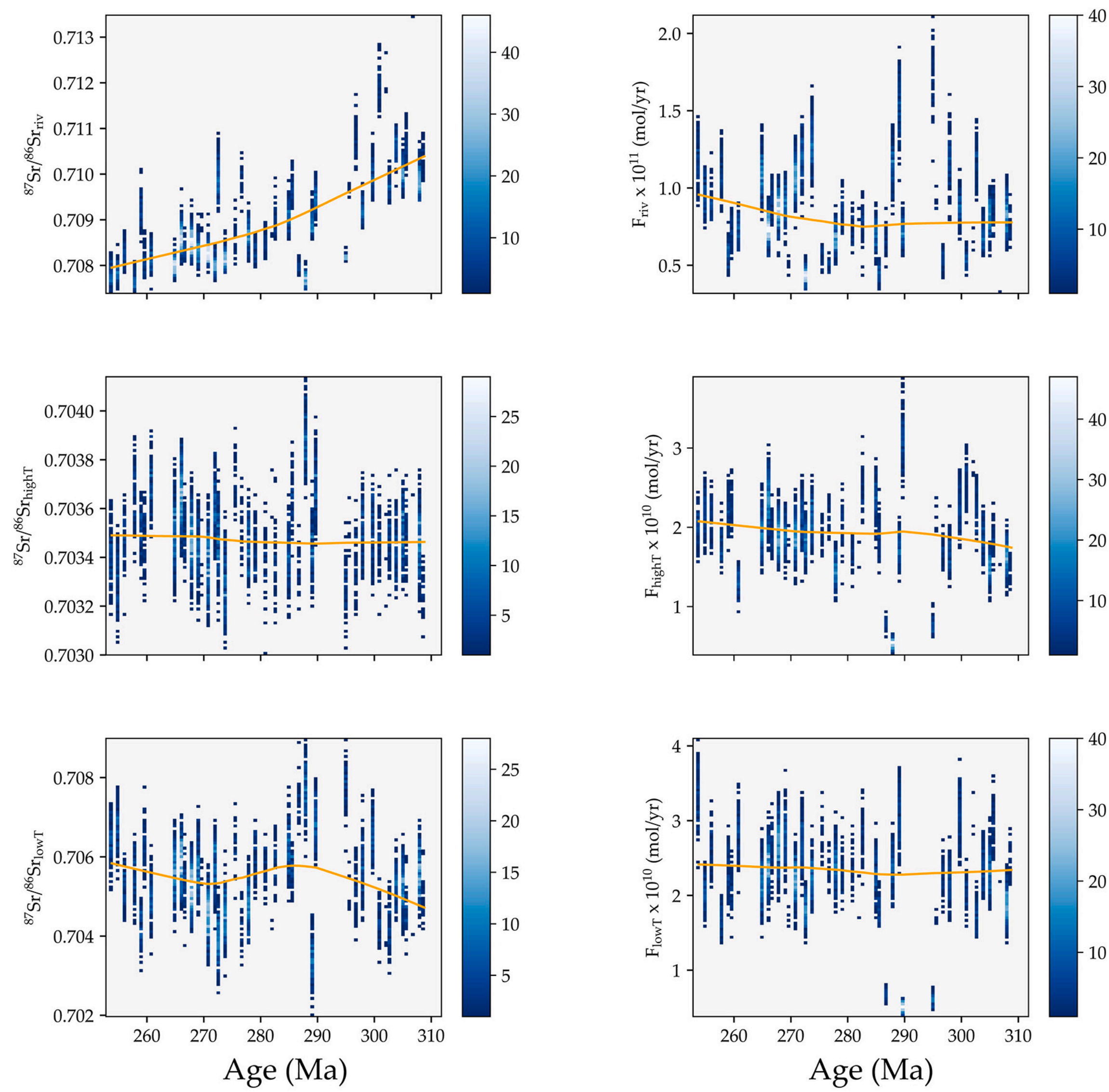

Fig. 10. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the diagenetic flux and isotopic composition (Scenario 1). Light blue colors indicate highest density (i.e., count) of successful parameters that match the empirically-determined, LOWESS-smoothed Sr isotope record using the 95% confidence intervals as filtering bounds for the stochastic model. Orange line is the LOWESS-smoothed mean of the density heatmap. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

temperatures likely increased from the Guadalupian to the Lopingian stages (Chen et al., 2013; Wang et al., 2020), we, therefore, allowed the low-temperature hydrothermal flux to vary by a factor of 4 relative to modern values (2.5–40 $\times$ 10^9 mol yr^{-1}) and the associated isotopic composition to vary between 0.7025 and 0.7084, representing a higher proportion of basalt alteration due to higher bottom water temperatures and approximately modern values, respectively. Investigating these large parameter spaces reduces bias inherent in model tuning and explores potential combinations of forcings on the Sr isotope system that may have been previously underexplored.

4. Results

4.1. Shell microstructure

The analyzed brachiopod shells belonging to the Orders Spiriferida and Athyridida contain fibrous secondary layers (Fig. 3A, B) and columnar tertiary layers (Fig. 3C). Specimens belonging to the Order Productida have laminar secondary layers (Fig. 3D) containing columnar tertiary layers. In the Bianping section, all the specimens are spiriferids with thick shells (about 4 mm), and most of them have a well-defined fibrous ultrastructure (Fig. 3B) and columns. In the Xikou section, shells with fibers and laminae were observed, and altered fibers were amalgamated (Fig. 3E). Altered columns show coarse surfaces (Fig. 3F). Moreover, the laminae of thick shells from the Shuixiakou Formation are morphologically well defined; instead, thin shells from other formations bear poorly defined laminae. In addition, the fibrous layers show better preservation compared to the laminar ones, because of the texture and size of the structural units (Garbelli et al., 2014).

Most of the shells ($n = 170$) analyzed are non-luminescent (Fig. 4A, B), with 45 shells showing slight luminescence (Fig. 4E) and 38 shells with outright luminescence (Fig. 4F). The secondary layers are luminescent more often than the tertiary layers (Fig. 4C). Some shells showed non-luminescent characteristics with bright luminescent fractures (Fig. 4D).

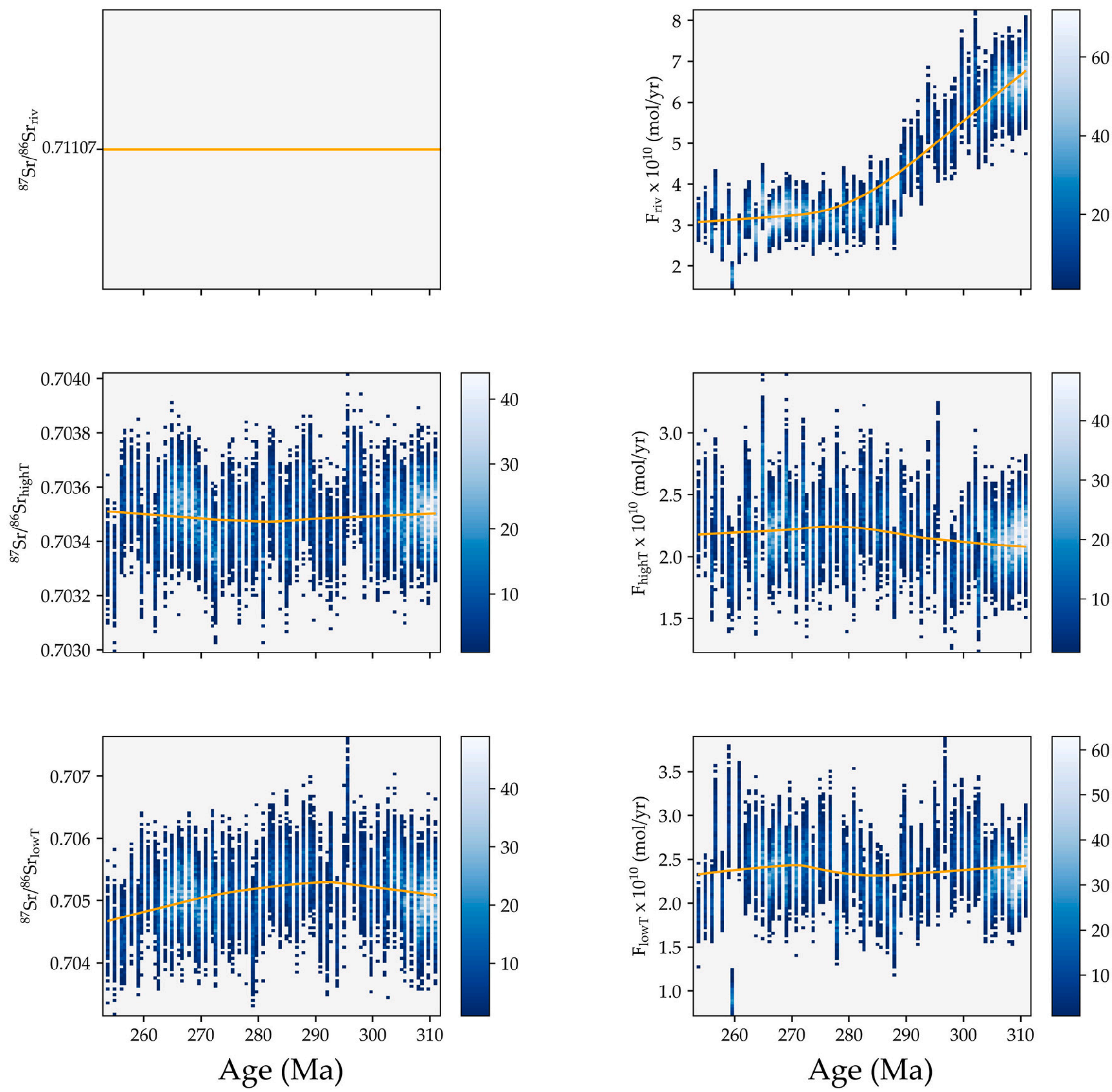

Fig. 11. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the riverine isotopic composition and diagenetic flux and isotopic composition (Scenario 2). See Fig. 2 for description.

4.2. Trace-elemental concentrations and carbon- and oxygen-isotope values

The range and distribution of Mg, Fe, Mn, and Sr for the Cisuralian samples are illustrated in Figs. 5 and 6, and the Guadalupian and Lopingian samples are all from the Xikou section, with different formations showing different covariance between Fe and Mn (Wang et al., 2020). Manganese contents of bulk rocks and brachiopod shells are both lower than 100 ppm for the Cisuralian samples, with most brachiopod shells being lower than 10 ppm (Fig. 5). There is overlap in Fe contents between brachiopod shells and bulk rocks from the Xikou and Bianping sections, while in the Shangsi section, Fe contents of bulk rocks samples are higher than that of brachiopod shells. For the Gudalupian and Lopingian samples, the highest values and widest range of these elements were recorded in bulk rocks, irrespective of the brachiopod shells' preservation state as based on SEM and cathodoluminescent microscopy observation (see Fig. 5 in Wang et al., 2020). As for the bulk rock samples, preserved shells, and altered shells, their geochemical properties overlap better with those in the Shuixiakou Formation compared with that in the Xikou and Yundoutan formations (Wang et al., 2020). In the Longdongchuan Formation there is no evident covariance between Fe and Mn, and the ranges of Fe overlap among all different materials (Wang et al., 2020).

The spiriferids in Shangsi, Bianping, and Xikou sections contain Mg 583–1055 ppm, 293–1379 ppm, and 177–4619 ppm, respectively. Samples from the Shangsi and Bianping sections contain a range of Mg concentration that is similar with previous results, while samples of the Xikou section bear a much broader range of Mg concentrations. Spiriferids in the Shangsi and Bianping sections bear Sr contents 434–719 ppm, 82–559 ppm, respectively. In the Xikou section, spiriferids contain Sr 175–660 ppm, with one sample yielding 1591 ppm as an outlier. Arthyridids contain 600–2302 ppm Mg, 1–178 Mn, 288–872 ppm Sr. Productids have much higher Mg (mean value, 3078 ppm) and Sr (mean value, 727 ppm, with most of them higher than 600 ppm) concentrations. The cross plot of Mg and Sr concentrations of Cisuralian samples shows that the brachiopod shells and their host rocks cluster in spatially separated fields all three sections, with only one shell from the Bianping section having Mg and Sr concentrations similar to that of the bulk sedimentary matrix (Fig. 6). For the Guadalupian and Lopingian samples in the Shuixiakou and Xikou formations, the ranges of Sr concentrations are smaller in brachiopod low-Mg calcite than in bulk rocks (Wang et al., 2020). Magnesium concentrations are higher in sedimentary matrix samples compared to brachiopod low-Mg calcite. For the Yundoutan and Longdongchuan formations (Wang et al., 2020), Mg and Sr cross plots

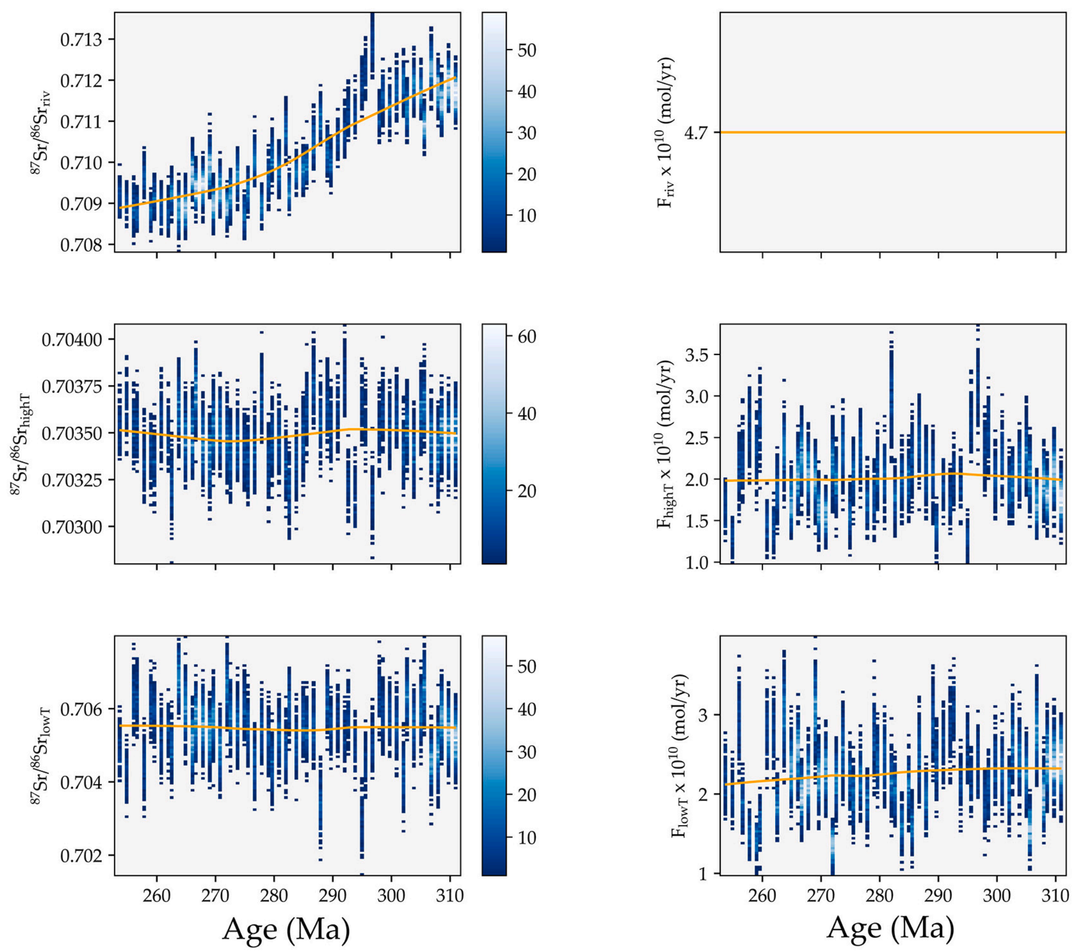

Fig. 12. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the riverine flux and diagenetic flux and isotopic composition (Scenario 3). See Fig. 2 for description.

show that the bulk rock samples are clearly separately clustered from the brachiopod low-Mg calcite (Wang et al., 2020).

The $\delta^{13}C$ and $\delta^{18}O$ values of Cisuralian samples are illustrated in Fig. 7. Generally, brachiopod shells have higher $\delta^{13}C$ and $\delta^{18}O$ values than that of whole rocks in all three sections, but the values of $\delta^{13}C$ and $\delta^{18}O$ overlap more in the Bianping section compared to the other two sections. The partially preserved brachiopod shells in the Xikou section tend to have lower $\delta^{18}O$ values than preserved ones, while the partially preserved shells have similar $\delta^{13}C$ values and $\delta^{18}O$ values with the preserved ones in the Shangsi section. Guadalupian and Lopingian samples show different $\delta^{13}C$ and $\delta^{18}O$ patterns in different formations (Wang et al., 2020). In the Shuixiakou Formation, the partially preserved brachiopod shells have lower $\delta^{13}C$ and $\delta^{18}O$ ($3.8 \pm 0.9‰$, $-3.7 \pm 0.5‰$), while the altered brachiopod shells and bulk rocks show similar $\delta^{13}C$ but lower $\delta^{18}O$ values (Wang et al., 2020). The partially preserved shells have similar $\delta^{13}C$ but lower $\delta^{18}O$ values than the preserved shells, whereas the enclosing rocks yield lower $\delta^{13}C$ and $\delta^{18}O$ values in the Xikou Formation (Wang et al., 2020). All carbonate materials in the Yundoutan Formation have similar $\delta^{13}C$ values, but the values of $\delta^{18}O$ decrease from preserved shells, partially preserved shells, altered shells to enclosing rocks, with the enclosing rocks have the lowest $\delta^{18}O$ values (Wang et al., 2020). In the Longdongchuan Formation, the average values of both $\delta^{13}C$ and $\delta^{18}O$ decrease from preserved shells, partially preserved shells, and altered shells to enclosing rocks, showing progressive diagenesis.

4.3. *Permian $^{87}Sr/^{86}Sr$ seawater data*

The range of $^{87}Sr/^{86}Sr$ ratios derived from well-preserved brachiopod shells is between 0.706832 and 0.708271 (Fig. 8). When plotted stratigraphically, we identify four main periods in the Sr isotope record: First, the Permian $^{87}Sr/^{86}Sr$ curve decreases rapidly from the Asselian to the end of the Kungurian stage (~ 299–273 Ma), with a rate of 0.000053 Myr^{-1}. Subsequently, the $^{87}Sr/^{86}Sr$ moderately decreases until the end of Wordian (middle Guadalupian) stage, with a rate of 0.000002 Myr^{-1}. From the beginning of the Capitanian to the middle Capitanian stages, the rate of change increases again, with a shift towards unradiogenic values at a rate of 0.000045 Myr^{-1} (Fig. 8). The lowest $^{87}Sr/^{86}Sr$ ratios of 0.706832 are registered in the *Colania douvillei-Kahlerina pachiytheca* Zone (early to middle Capitanian). Afterward, the $^{87}Sr/^{86}Sr$ values increase until the latest Changhsingian stage and reach values of 0.707167, just 0.8 m below the first occurrence of the *H. parvus*. The increasing trend of the Lopingian is characterized by two intervals, the gently increasing interval during the Wuchiapingian (0.000014 Myr^{-1}) and the rapidly increasing interval during the Changhsingian stage (0.000076 Myr^{-1}; Fig. 8).

5. Discussion

5.1. *Diagenetic evaluation of brachiopod shells*

Given that diagenesis will alter the primary $^{87}Sr/^{86}Sr$ signal, it is fundamental to evaluate the preservation state of materials used for seawater $^{87}Sr/^{86}Sr$ reconstruction (Brand et al., 2012a; Korte et al.,

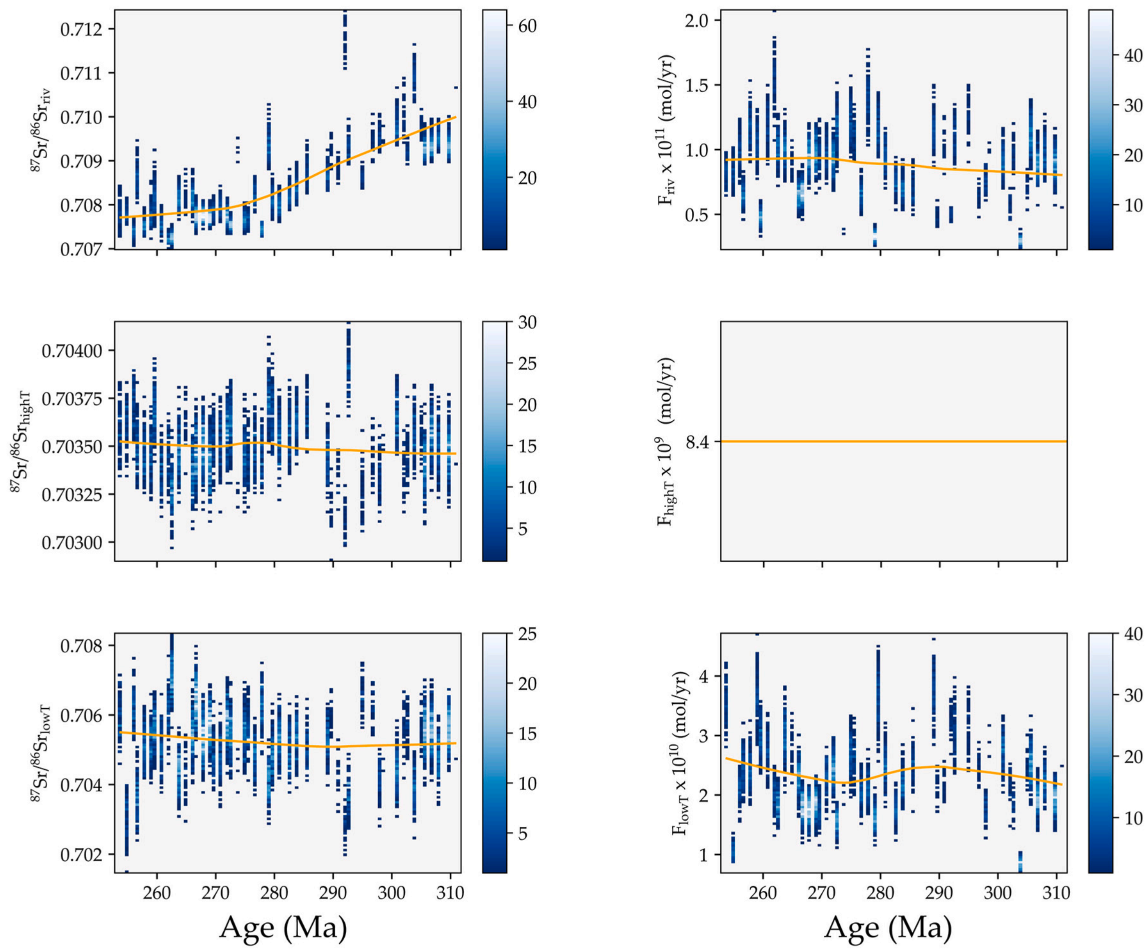

Fig. 13. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the high-temperature hydrothermal flux and diagenetic flux and isotopic composition (Scenario 4). See Fig. 2 for description.

2006; McArthur et al., 2012; Immenhauser et al., 2016; Wang et al., 2018). Morphological examinations under SEM and cathodoluminescent microscopy were performed as the first screening step (Brand et al., 2012a; Ullmann and Korte, 2015). Any texturally altered shells were eliminated from the subsequent analyses. The major and trace elemental as well as carbon and oxygen isotope data were analyzed to further distinguish the preservation state of shells.

The microstructure of the shells is a crucial factor in determining the preservation state of the brachiopod shells. The Cisuralian brachiopods are from the Bianping and Shangsi sections. A total of 96% ($n = 49$) of shells from the Bianping section and 73% ($n = 8$) of the shells from the Shangsi section show well-defined microstructure and non-luminescent characteristics, suggesting the brachiopods from the two sections did not experience significant alteration. For the Cisuralian brachiopods from the Xikou section, 42% (n = 8) of the shells were altered based on SEM and cathodoluminescent microscopy observation. Guadalupian and Lopingian samples are from four different formations in the Xikou section (Wang et al., 2020). The preservation state of the brachiopod shells from different formations at the Xikou section was discussed in Wang et al. (2020) and summarized in Table 2. The amount of altered shells positively correlated with shells containing laminar secondary layers, suggesting the fibrous layers are less susceptible to diagenetic alteration than their coeval laminar counterparts (Garbelli et al., 2012).

The elemental concentration (e.g., Sr, Mn, and Fe) of the brachiopod shells are widely used as diagenetic indicators since meteoric diagenesis likely leads to decreased Sr content and elevated Mn and Fe contents (Brand and Veizer, 1980; Garbelli et al., 2019; Wang et al., 2018). The ranges on well-preserved modern brachiopods are 450 to 1930 ppm for Sr, below 200 ppm for Mn, and below 140 ppm for Fe (Brand et al., 2003). Many publications use modern brachiopod-based ranges (e.g., Mn < 250 ppm, Sr > 400 ppm; Korte et al., 2006), as the cutoff for diagenetic resetting. Brand et al. (2011) suggested dynamic limits for trace-elemental concentrations to differentiate preserved and altered shell materials. Recent research showed that well-preserved Rhynchonellata have lower Mg (834 to 1347 ppm), Sr (256 to 423 ppm), and Fe concentrations than those of the Strophomenata (Garbelli et al., 2014). Our data show that productids were enriched in Sr and Mg. Overall, it is essential to know the taxonomic information before using general cutoffs of certain elemental concentrations diagenesis-screening. In the present study, for spiriferids and arthyridids, Sr concentrations that are less than 250 ppm, higher than 600 ppm are considered as altered, while for productids, Sr content less than 500 ppm, higher than 1000 ppm are considered altered, since those with higher and less Sr both correlate with higher $^{87}Sr/^{86}Sr$ (Appendix B). Iron and Mn contents, on the other hand, do not show obvious species-dependency. Our study documents that most preserved brachiopod shells have Mn contents lower than 50 ppm, and many of them are lower than 10 ppm (Appendix B), suggesting an overall good preservation.

In summary, only brachiopod shells with a well-preserved microstructure, no luminescence, and with low Mn (< 50 ppm) and high Sr (spiriferids and arthyridids with Sr concentrations between 250 ppm and 600 ppm, productids with Sr concentrations between 500 ppm and 1000 ppm) contents are considered well-preserved in this study and discussed in the following sections.

Finally, the $^{87}Sr/^{86}Sr$ ratio of the matrix sedimentary rocks, to some extent, could be a useful indicator of alteration. Given the residence time

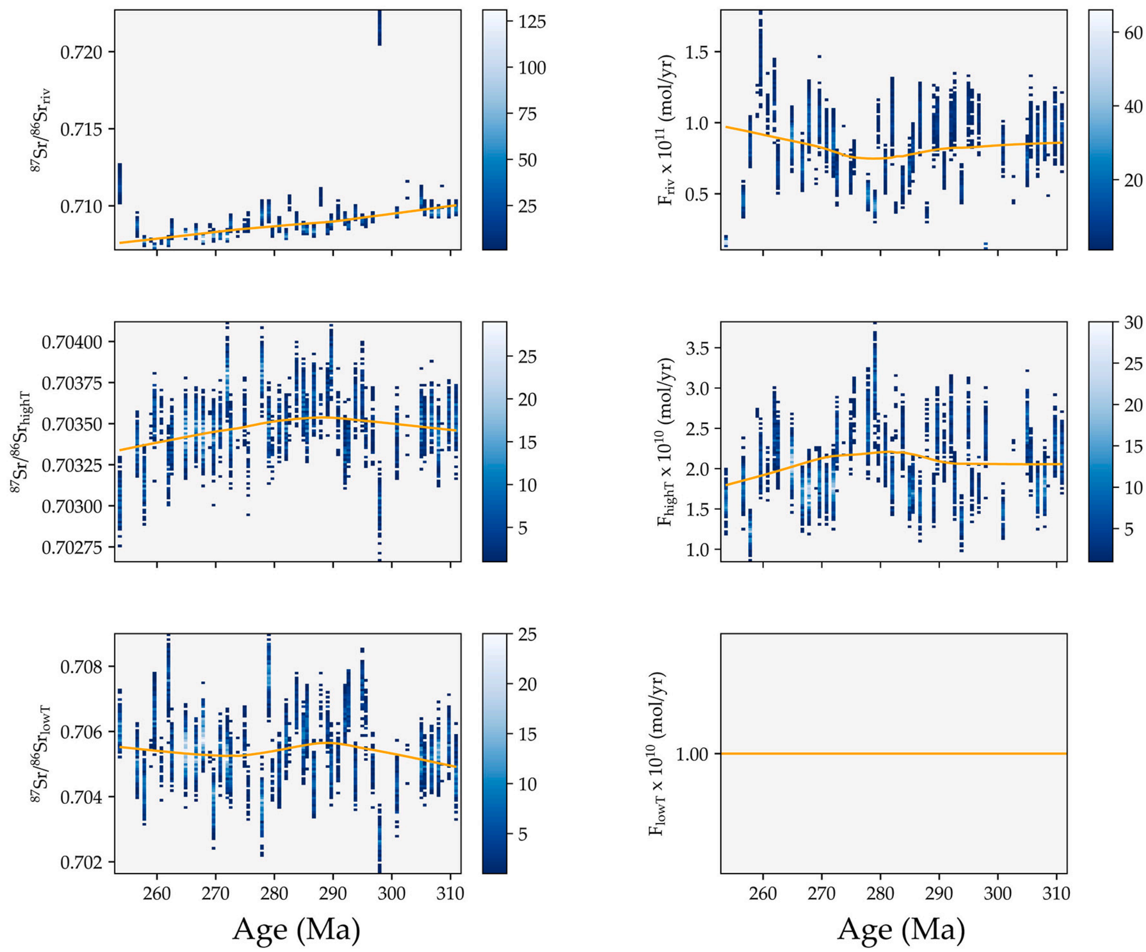

Fig. 14. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the low-temperature hydrothermal flux and diagenetic flux and isotopic composition (Scenario 5). See Fig. 2 for description.

of Sr, we applied the $^{87}Sr/^{86}Sr$ fluctuation in modern biogenetic calcite (± 0.000054; Zaky et al., 2019) to ancient carbonates within 1 Myr intervals and exclude those anomalous high $^{87}Sr/^{86}Sr$ ratios (relative to their coeval counterparts) in our discussion.

5.2. Comparison with the previously published seawater $^{87}Sr/^{86}Sr$ curve

The Permian $^{87}Sr/^{86}Sr$ curve is characterized by the long-term decrease from the earliest Cisuralian to the Capitanian stages, recording the lowest Sr isotope values during the Phanerozoic followed by an increase to more radiogenic values at the end of the Capitanian stage through the PTB (Denison et al., 1994; Korte et al., 2006; McArthur et al., 2012; Morante, 1996; Song et al., 2015; Wang et al., 2018). The general trend in this study is consistent with previous studies (Korte et al., 2006; McArthur et al., 2020; Wang et al., 2018). Given that we present a higher resolution record, we capture some major differences that merit discussion.

First, the rate of change in $^{87}Sr/^{86}Sr$ ratios differs from previous studies. Previous records show a rapid decrease from the Asselian to Wordian stages (McArthur et al., 2020). While our $^{87}Sr/^{86}Sr$ ratios decrease rapidly from the earliest Asselian to the Kungurian stages, followed by a gentle decrease from the Kungurian to Woridan stages (Fig. 8). We tend to interpret this inconsistency as the different $^{87}Sr/^{86}Sr$ ratios of the Kungurian and the early Capitanian, considering the discrepancy of $^{87}Sr/^{86}Sr$ ratio for the Carboniferous-Permian boundary between our study and previous studies are within the fluctuation of modern calcitic brachiopods (Chernykh et al., 2016; Korte et al., 2006; Zaky et al., 2019). In our study, the Kungurian brachiopods are collected in the Chihsia Formation at the Shangsi section. Based on our diagenesis-screening criteria, the samples from the Shangsi section are texturally and geochemically well-preserved, yet their $^{87}Sr/^{86}Sr$ ratios are consistently lower than the data from Korte et al. (2006). We note that the "good" Kungurian brachiopods $^{87}Sr/^{86}Sr$ data in Korte et al. (2006) contain only two points obtained from the Guadalupe Mountains with no biozone constraints. In contrast, our Kungurian brachiopod shells may have better age constraints because of the concurrent conodont elements *Pseudosweetognathus costatus* and *Mesogondolella lamberti*. Similarly, the $^{87}Sr/^{86}Sr$ records of the brachiopod shells tied to palynology zones from the Bowen Basin in eastern Australia (Morante, 1996) exhibit similar values as our records, despite the fact that the palynology zones may not correlate globally. As for the Capitanian, our samples are from the Xikou section that concurrent with fusulinid *Chenella changanchiaoensis*, while the LOWESS V6 (McArthur et al., 2020) compiled the data from Kani et al. (2018) and Garbelli et al. (2019), all of which are not tied with biostratigraphy that can be correlate globally. Therefore, $^{87}Sr/^{86}Sr$ ratios tied with biostratigraphy markers of global significance are of crucial relevance when aiming to establish a global Sr isotope stratigraphy (Korte and Ullmann, 2016; Wang et al., 2018).

Second, the temporal position of the lowest $^{87}Sr/^{86}Sr$ ratios are better constrained in this study. The lowest $^{87}Sr/^{86}Sr$ ratios during the Permian (Denison and Koepnick, 1995; Korte et al., 2006; McArthur et al., 2012) are widely used as an important datum (Kani et al., 2008, 2013, 2018; Morante, 1996). Korte et al. (2006) reported the lowest $^{87}Sr/^{86}Sr$ ratio of 0.706854 based on the well-preserved brachiopod shells at the beginning of the Capitanian, but there is another minimum (0.706852) close to the GLB. Wang et al. (2018) analyzed the micritic limestone at the

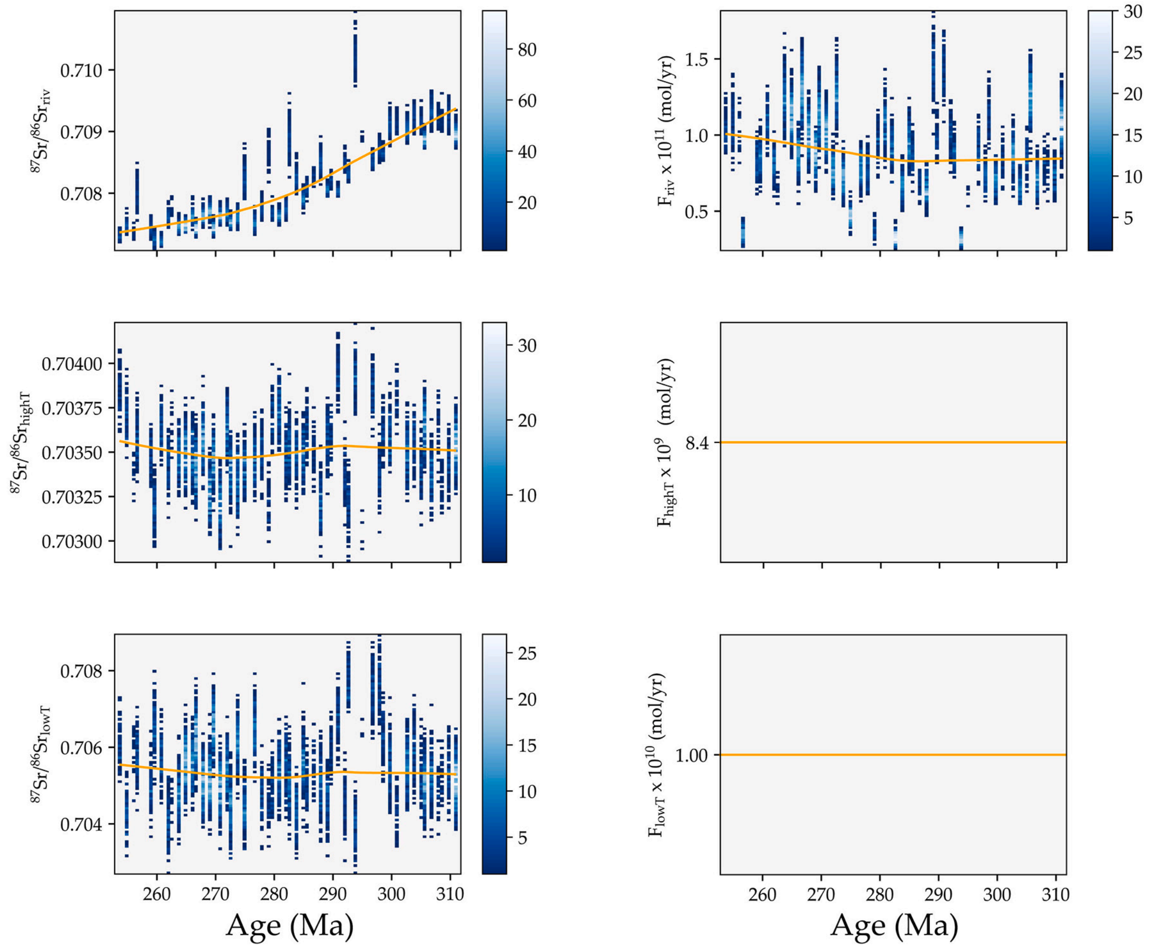

Fig. 15. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding the high- and low-temperature hydrothermal fluxes and diagenetic flux and isotopic composition (Scenario 6). See Fig. 2 for description.

GSSP of the base of the Lopingian – the Penglaitan section – with the lowest value registered within the conodont zone *Jinogondolella shannoni/J. altudaensis* and *J. xuanhanensis* (late Capitanian). The lowest $^{87}Sr/^{86}Sr$ ratio recorded by the micritic limestone, however, is higher than that of brachiopod shells (Korte et al., 2006). The lowest $^{87}Sr/^{86}Sr$ ratio in our study is 0.706838, but the indistinguishably low value of 0.706863 is recorded 2.85 Myr before this lowest value (Appendix B). Thus, the lowest values in our study were constrained within the fusulinid *Colania douvillei-Kahlerina pachytheca* Zone lasting for 2.85 Myr.

Third, the $^{87}Sr/^{86}Sr$ ratios were better constrained during the Changhsingian stage. The increasing rate of change in $^{87}Sr/^{86}Sr$ ratios during the Changhsingian was estimated about 0.000109 Myr^{-1} by Korte and Ullmann (2016) and was interpreted as evidence for higher weathering rates during the latest Permian period triggered by the eruption of Siberian Traps Igneous Province (Brand et al., 2012b; Cao et al., 2009; Dudás et al., 2017; Garbelli et al., 2016; Huang et al., 2008; Korte et al., 2010; Liu et al., 2013; Schobben et al., 2014; Sedlacek et al., 2014; Song et al., 2015; Wang et al., 2018).

The brachiopod-shell derived $^{87}Sr/^{86}Sr$ ratios obtained near the PTB interval in South China is 0.70715 in Korte et al. (2006), which is lower than that of host-rock samples (Cao et al., 2009; Sedlacek et al., 2014) but similar to the brachiopod shells from Italy (Brand et al., 2012b) and Tibet, China (Garbelli et al., 2016). The high-resolution $^{87}Sr/^{86}Sr$ ratios obtained from conodonts in Dudás et al. (2017) yield results of 0.70708 around the PTB interval. Therefore, it is important to exclude diagenetically reset samples and their corresponding data when interpreting the marine $^{87}Sr/^{86}Sr$ records.

5.3. Model discussion and controlling factors of the seawater $^{87}Sr/^{86}Sr$ evolution

We investigated several scenarios to evaluate the potential drivers of the Permian Sr isotope record, all of which can determine adequate combinations of fluxes and isotopic compositions to match the empirical record (Fig. 9). First, given that the strontium system may have been different relative to modern, we allow six out of eight parameters (the diagenetic flux and isotopic composition are assumed as constant) to vary by a large range, which are filtered by solely keeping solutions to the steady-state Sr isotope mass balance that match the empirical record (Fig. 10). The second scenario (Fig. 11) keeps the isotopic composition of the riverine input constant and set to the modern value, assuming that there is no change in the riverine isotopic signature due to the evolution of the weathered rocks at Earth's surface over this time interval (*c.f.* Bluth and Kump, 1991). The third scenario (Fig. 12) follows a similar reasoning but evaluates the model's sensitivity by setting the riverine flux constant. The fourth, fifth, and sixth scenarios (Figs. 13-15) set the high- and low-temperature hydrothermal fluxes equal to modern throughout the entire model run, investigating the potential solution space given no changes in the predominantly unradiogenic fluxes of Sr. Lastly, the seventh scenario keeps the riverine flux and isotopic composition constant to evaluate the impact of shifts in high- and low-temperature hydrothermal systems.

Scenarios 1 (Fig. 10), 4 (Fig. 13), 5 (Fig. 14), and 6 (Fig. 15) all indicate increases in the riverine flux coupled with significant decreases in the riverine isotopic composition. The remaining parameters remain relatively constant, barring Scenario 1 showing the low-temperature

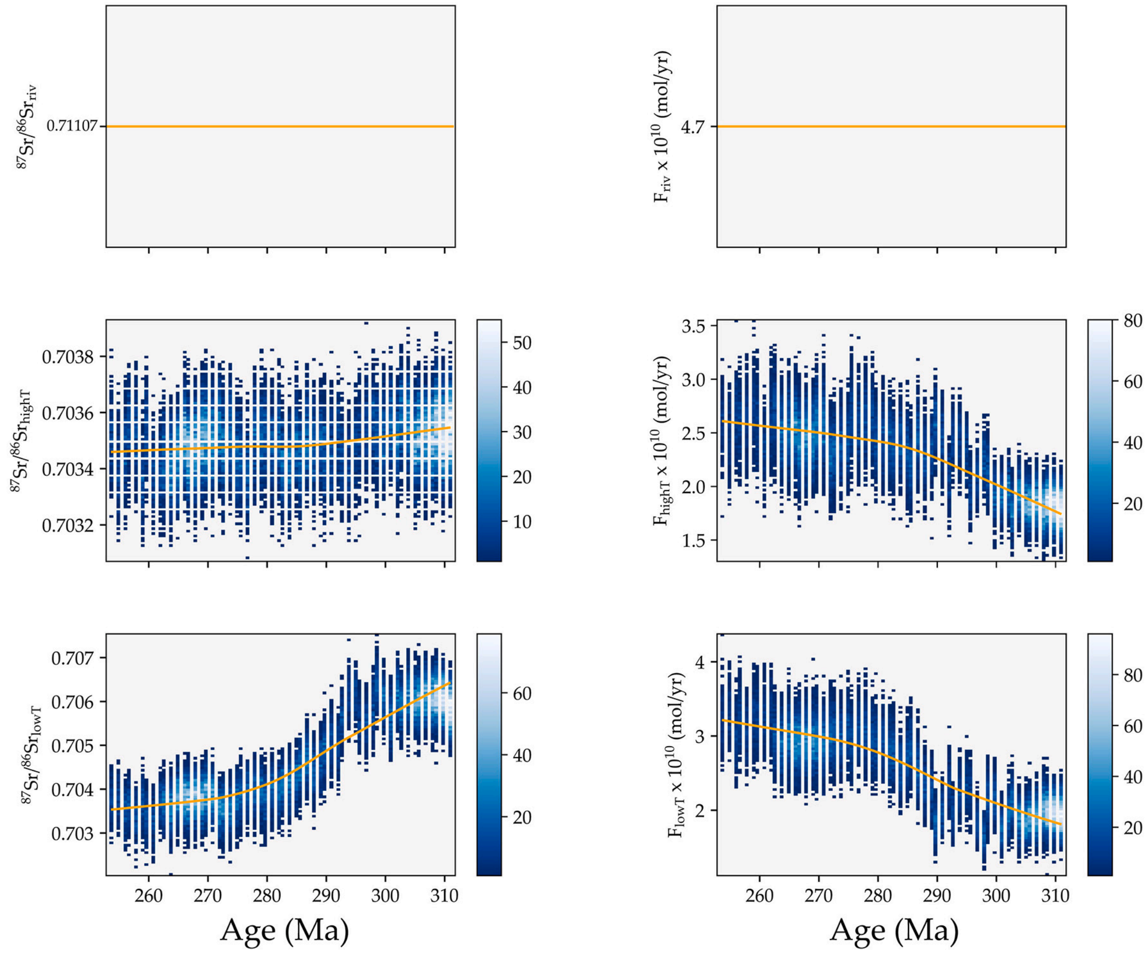

Fig. 16. Two-dimensional density heatmap of seawater strontium isotope mass balance results with all parameters varying, excluding riverine flux and isotopic composition, and diagenetic flux and isotopic composition (Scenario 7). See figure10 for description.

hydrothermal isotopic composition increasing to more radiogenic values. Scenario 2 (Fig. 11), which holds the riverine isotopic composition constant, compensates with a significant decrease in the riverine flux, and slight increases in the high-temperature hydrothermal flux and the low-temperature hydrothermal isotopic composition. Similarly, scenario 3 (Fig. 12) requires a large shift in the riverine isotopic composition to unradiogenic values with the riverine flux set to the modern value. All other parameters remain relatively stable for this scenario. Scenario 7 (Fig. 16) requires large increases in high- and low-temperature hydrothermal fluxes, accompanied with a shift in the low-temperature hydrothermal isotopic composition to unradiogenic values.

The Permian Sr record, which captures a dramatic shift to unradiogenic values, is coincident with various paleoenvironmental perturbations such as the building and melting of ice shields (Korte et al., 2006; Kani et al., 2013; Montañez and Poulsen, 2013; Chen et al., 2016), rifting leading to the opening of the Neotethys (Korte et al., 2006; Wang et al., 2018), and global warming (Chen et al., 2013; Wang et al., 2020). Each of these shifts in Earth's system state was likely to have led to a corresponding perturbation to the Sr isotope system. Our model results can provide insight to the most plausible explanatory mechanism. In the majority of scenarios presented, the dramatic shift to unradiogenic values captured by the empirical Sr isotope record involves a perturbation to the terrestrial Sr input.

Scenarios 1–6 require either a shift in the flux or isotopic composition of riverine Sr. Unless both of these parameters are held constant (Fig. 8), it seems most likely that changes in continental weathering were required to match the empirical record. Glaciation, deglaciation, and Euramerica aridification may have been the culprit for changes in the Sr riverine flux (Chen et al., 2018; Korte et al., 2006; Wang et al., 2018). Given the long duration of the decrease in the Sr record, however, it is unlikely that the effects of glaciation during the Cisuralian (Montañez and Poulsen, 2013) are responsible for the entirety of the unradiogenic trend (Wang et al., 2018). Furthermore, deglaciation should expose significant amounts of freshly eroded bedrock (Blum and Erel, 1995) and glacial loess which is likely to have a relatively radiogenic signature (Colin et al., 1999, 2006), and yet, the continuously decreasing Sr record indicates that the behavior of the empirical record may be unrelated to enhanced weathering.

Rather than invoking changes in the riverine flux, the decrease in the Sr record may be related to shifts in the composition of weathering of rocks at Earth's surface given that different lithologies may provide a weathering signal with different isotopic compositions (Palmer and Edmond, 1992). Bluth and Kump (1991) estimated a long-term decrease in the relative exposure area of high $^{87}Sr/^{86}Sr$ shield rocks (Palmer and Edmond, 1992). Although this decrease is consistent with the scenarios that show the riverine isotopic composition decreasing over this time interval (Figs. 10, 12–15), the change in exposure area of certain rock types is sustained for periods of time much longer than the Permian and therefore does not provide a direct explanation for the Permian Sr isotope record.

Enhanced hydrothermal activity, as a consequence of the opening of the Neotethys, has been suggested as the main driver of the increasingly unradiogenic Permian trend (Korte et al., 2006; Wang et al., 2018). In scenario 7 (Fig. 16), this effect is most obvious when the riverine flux and isotopic composition are kept constant, suggesting that increases in the hydrothermal flux (from both high- and low-temperature environments) are required. Furthermore, scenario 7 indicates that the isotopic composition of the low-temperature hydrothermal flux is likely to have decreased, consistent with the general warming trend across the Permian (Chen et al., 2013) leading to potentially more effective leaching of unradiogenic basalt in low-temperature, off-axis hydrothermal systems (e.g., Coogan and Dosso, 2015).

Although high-temperature hydrothermal fluid input likely had an effect on the seawater chemistry fluctuations over this time period (Antonelli et al., 2017), this effect was unlikely to be the main driver of the Sr isotope record with the high-temperature hydrothermal isotopic composition decreasing ~0.001 (from ~0.704 to 0.703) across the Permian. This unradiogenic shift occurred in the early Permian and was short-lived with R_{highT} rebounding to more radiogenic values in the mid- to late Permian. Therefore, this explanatory mechanism is likely insufficient for driving the empirical record.

In summary, our model indicates that there are many potential solutions to the empirical Sr isotope record for the Permian. The majority of our model scenarios point to major shifts in riverine isotopic compositions, suggesting that changes in the material being weathered at a global scale may have shifted across the Permian. However, the more likely scenario, given significant tectonic activity occurring during the Permian associated with opening of the Neotethys and dispersal of Pangea, is that enhanced unradiogenic input from hydrothermal systems may have been the main driver of the empirical Sr record. In addition to potentially increased hydrothermal activity, increasingly warming temperatures throughout the Permian may have led to enhanced low-temperature basalt alteration, shifting the isotopic composition of fluids emanating from off-axis hydrothermal systems to more unradiogenic values. Therefore, we suggest that perturbations to Earth's hydrothermal systems may have been the likely drivers of pronounced unradiogenic trend across the Permian.

6. Conclusions

In this paper, we review the previously published Permian $^{87}Sr/^{86}Sr$ record, present a new high-resolution brachiopod-based $^{87}Sr/^{86}Sr$ curve, and discuss the possible controlling factors using a stochastic oceanic box model. The new $^{87}Sr/^{86}Sr$ record is generally consistent with previous results, with discrepancies present at higher resolution. Our record shows that $^{87}Sr/^{86}Sr$ ratios decrease from the earliest Permian to middle Capitanian, with the lowest values registered in the fusulinid *Colania douvillei-Kahlerina pachiytheca* Zone. Following this interval, $^{87}Sr/^{86}Sr$ ratios increase from the Capitanian to the latest Permian and reach 0.707167 just 0.8 m below the first occurrence of the *H. parvus*. The model results suggest that change in unradiogenic Sr input from the hydrothermal systems is the most likely control on the Permian seawater $^{87}Sr/^{86}Sr$ ratio and its temporal trends.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Professor John M. McArthur is acknowledged for providing us the latest version of Sr look-up table. Two anonymous reviewers are thanked for their constructive suggestions. This study is supported by the Strategic Priority Research Program (B) of the Chinese Academy of Sciences (XDB18000000, XDB26000000), the Natural Science Foundation of China (42073002, 91955201, 41830323, 42072013), and CAS (QYZDY-SSW-DQC023) for SZS, WQW, CG, and DXY.

Appendix. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.earscirev.2021.103679.

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